IECE 672 – Foundations of Statistical Inference – Spring 2026

Project Description

Due Date: May 5th, 2026 (11:59pm)

What to Submit: Your final submission should happen via Brightspace and include a single compressed file (i.e., `LastName_IECE672_Project.zip`) with the following files: (i) a PDF file with your report in the template provided in Brightspace (`Project_Report_Template.zip`), and (ii) your Python or Matlab code (properly commented). Make sure to include your full name and student id at the beginning of your report.

Part 1. SeeClickFix[1] constitutes an easy–to–use Web interface, which assists concerned citizens report problems to governmental agencies regarding their local environment. More specifically, users can describe 311 issues (e.g., parking violations, snow removal, animal control, etc.) by submitting an online request that includes information such as title, description, category, image, and location. A SeeClickFix dataset that includes requests in the Albany County region (`SeeClickFix_AlbanyCounty_February_2018.csv`) has become available to you on Brightspace. Each row in the dataset corresponds to a single request described by the tuple (`issue_id`, `feature 1`, ..., `feature K`, `category`), where `issue_id` corresponds to a unique identification number given to each request, `feature` i denotes the number of times feature i appeared in the corresponding request, and `category` can be one of 'Parking Enforcement', 'Code Violations', 'Signs (Missing, Needed, or Damaged)' and 'Traffic Signal Repairs'. The features correspond to $2,535$ unique words (after preprocessing) extracted from the title and description of 529 physical requests. We assume that the features are independent under each of the four categories.

Task 1. Binary Hypothesis Testing under the Bayes Criterion. For each of the scenarios below, design a binary hypothesis test to classify requests between: (a) 'Parking Enforcement' and 'Code Violations', and (b) 'Signs (Missing, Needed, or Damaged)' and 'Traffic Signal Repairs'. Numerically evaluate and report the values of P_F, P_D, P_M, and P_ϵ as a function of $C_{00}, C_{01}, C_{10}, C_{11}$ in table format. Draw the associated ROC in eah case and discuss your observations. To estimate the parameters required to evaluate your approach, split your dataset into training and testing, so that half of the requests in a particular category are in the training dataset and the rest are in the testing dataset. Your report should include (i) a detailed description of your approach including key equations and related derivations, (ii) a description of your evaluation procedure, and (iii) a presentation and discussion of your results.

- **Multi–variate Bernoulli Model.** The multi–variate Bernoulli model assumes that every feature is associated with the value 1 or 0; the value 1 indicates that the word occurs in the particular request, and 0 indicates that the word does not occur in the particular request. For a feature vector $\mathbf{x}$ of size m, where m is the number of words in the dataset, the conditional probability density can be written as:

$$P(\mathbf{x}|H_j) = \prod_{i=1}^{m} P(x_i|H_j)^b(1 - P(x_i|H_j))^{1-b}, \quad b = 0, 1, j = 0, 1. \tag{1}$$

[1] https://seeclickfix.com/

To evaluate the performance of your binary hypothesis test, you will need to compute the *a priori* probabilities $P_j, j = 0, 1$, and conditional probabilities $P(x_i|H_j), j = 0, 1$. To this end, use the training dataset to compute the maximum likelihood estimate of $P_j, j = 0, 1$, and estimate $P(x_i|H_j), j = 0, 1$, using the following maximum likelihood estimate:

$$\hat{P}(x_i|H_j) = \frac{N_{ij} + 1}{Nj + 2}, \tag{2}$$

where N_{ij} is the number of requests in the training dataset that contain feature x_i and belong to hypothesis H_j and N_j is the number of requests in the training dataset that belong to hypothesis H_j.

- **Multinomial Model.** An alternative approach to characterize the requests is the term frequency, i.e., the number of times a given word appears in a given request. In practice, the term frequency is often normalized by dividing the raw term frequency by the document length. To evaluate the performance of your binary hypothesis test in this case, estimate $P(x_i|H_j), j = 0, 1$, from the training dataset using the following maximum likelihood estimate:

$$\hat{P}(x_i|H_j) = \frac{\sum tf_{ij} + 1}{\sum N_j + V}, \tag{3}$$

where $\sum tf_{ij}$ is the sum of raw term frequencies of word x_i from all requests in the training sample that belong to hypothesis H_j, $\sum N_j$ is the sum of all term frequencies in the training dataset for hypothesis H_j and V is the number of different words in the training dataset. The conditional probability of a feature vector $\mathbf{x}$ of size m is then:

$$P(\mathbf{x}|H_j) = \prod_{i=1}^{m} P(x_i|H_j), \quad j = 0, 1. \tag{4}$$

Task 2. Multiple Hypothesis Testing under the Bayes Criterion. Repeat all parts of Part 1 in the case where you wish to classify between the four hypotheses 'Parking Enforcement', 'Code Violations', 'Signs (Missing, Needed, or Damaged)' and 'Traffic Signal Repairs'.

Part 2. Download and carefully read the following paper:

Chao Huang, Dong Wang, "Unsupervised Interesting Places Discovery in Location–Based Social Sensing," 12th IEEE International Conference on Distributed Computing in Sensor Systems, 2016.

Task 1. Expectation–Maximization Algorithm. Explain the Expectation–Maximization Algorithm. Your report should include a brief description of its goal, discussion of related equations, challenges and any other relevant observations. (Hint: Reference [6] can be very helpful, but feel free to explore other references as well).

Task 2. Paper Summary. Write a detailed summary of the above paper. Your report should include a description of: (i) the problem and its formulation (make sure to include key equations), (ii) the proposed solution, (iii) the evaluation protocol, and (iii) 1–3 important outcomes.

[Bonus (Optional)] **Task 3. Implementation.** Download one of the two publicly available datasets (Brightkite and Gowalla) mentioned in the paper and replicate the results shown in Figure 7 or 8 for algorithms PSIPD–P, PSIPD–S and PSIPD–PS.